carbon, and provide material to be decomposed by bacteria and fungi, in turn providing soil nutrients for plants.

Tansley also argued that these internal processes within an ecosystem conform to what he described as “the great universal law of equilibrium.” Self-regulating, these processes have a natural tendency toward stability. The cycles within an ecosystem contain feedback loops that correct any fluctuations from a state of equilibrium. Each ecosystem is located in a particular area, with characteristics unique to its environment, and behaves as a self-contained and self-regulating system. Together, the patchwork of ecosystems across the globe form what Austrian scientist Eduard Suess called the biosphere—the sum total of all ecosystems.

External factors

Various external factors, such as climate and the geological makeup of the surrounding environment, can affect an ecosystem. One constant external force affecting all ecosystems is the Sun. The energy it supplies enables photosynthesis and the capture of CO_2 from the atmosphere; some of this energy is distributed through the ecosystem and the food chain. In the process, some of this energy dissipates as heat.

Other external factors, however, can arise unexpectedly to create pressures on ecosystems. From time to time, all ecosystems are subjected to external disturbances, and must then go through a process of recovery. These disturbances include storms, earthquakes, floods, droughts, and other natural phenomena, but are increasingly the result of human activity—due to the destruction of natural habitats by deforestation, urbanization, pollution, and the cumulative effects of anthropogenic (human-induced) climate change. Humans can also be responsible for the introduction of invasive species. Without external factors such as these, an ecosystem would maintain a state of equilibrium, and retain a stable identity.